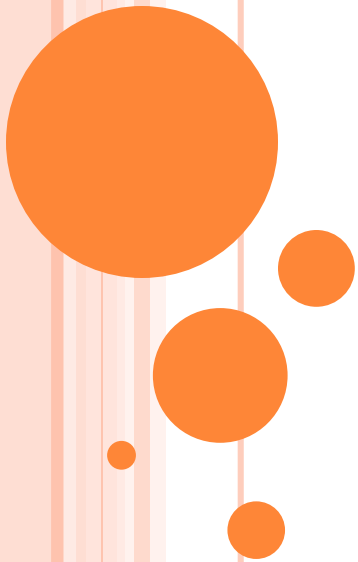


MCDM

MULTI-CRITERIA DECISION MAKING



○ Grading:

- Midterm Exam: 30%
- Final Exam: 40%
- Homeworks: 10%
- Project: 20%

○ Textbook:

- “Multiple Attribute Decision Making: Methods and applications”. Gwo-Hshiung Tzeng & Jih-Jeng Huang, CRC Press, Taylor & Francis Group, 2011 by Taylor & Francis Group.

CONTENTS:

- MCDM definition
- Problem solving steps
- Criteria specifications
- Weighting the criteria
- Standardizing the raw scores
- Problem solving techniques

MCDM DEFINITIONS

- Almost (if not all) decision making problems deal with multiple objectives/criteria.
- MCDM includes MODM and MADM:
 - MODM (Multiple Objective DM): Finding out the set of **non-dominated** solutions from infinite solutions, according to multiple conflicting **objectives**.
 - MADM (Multiple Attribute DM): selection of the best solution from a set of alternatives, each of which is evaluated against multiple conflicting **criteria**.
- **Conflicting property**: by nature, people consider conflicting objectives or conflicting criteria, if not, they can be combined.

MCDM DEFINITIONS

- **Goal:** set of target values which are expected to achieve. (ex: 1000 ton of material next year)
- **Objectives:** directions of achievements, improvements (ex: Maximize Profit)
- **Criteria:** A set of attributes used to evaluate alternatives (ex: Price will be used to evaluate a product to buy).
- **Factor:** A set of similar of attributes and/or actions grouped together to reflect some evaluation.
- **Alternatives:** A set of options (after screening)

EXAMPLES OF MULTI-CRITERIA PROBLEMS

- In determining an electric route for power transmission in a city, several criteria could be considered:
 - Cost
 - Health
 - Reliability
 - Importance of areas

EXAMPLES OF MULTI-CRITERIA PROBLEMS

- Locating a nuclear power plant involves criteria such as:
 - Safety
 - Health
 - Environment
 - Cost

PROBLEM SOLVING STEPS:

- 1) Establish the decision context, the decision objectives (goals), and identify the decision maker(s).
- 2) Identify the alternatives.
- 3) Identify the criteria (attributes) that are relevant to the decision problem.

Problem solving steps:

- 4) For each of the criteria, assign scores to measure the performance of the alternatives against each of these and construct an evaluation matrix (often called an options matrix or a decision table).

The structure of an options matrix (S)

Criteria	Alternatives				
	a_1	a_2	a_3	...	a_i
c_1	0.1				s_{i1}
c_2	0.3				s_{i2}
c_3	0.7				s_{i3}
...					...
c_j	s_{1j}	s_{2j}	s_{3j}	...	s_{ij}

PROBLEM SOLVING STEPS:

- 5) Standardize the raw scores to generate a priority scores matrix or decision table.
- 6) Determine a weight for each criterion to reflect how important it is to the overall decision.

Problem solving steps:

- 7) Use aggregation functions (also called decision rules) to compute an overall assessment measure for each decision alternative by combining the weights and priority scores.
- 8) Perform a sensitivity analysis to assess the robustness of the preference ranking to changes in the criteria scores and/or the assigned weights.

CRITERIA CHARACTERISTICS

- **Completeness:** It is important to ensure that all of the important criteria are included.
- **Redundancy:** In principle, criteria that have been judged relatively unimportant or to be duplicates should be removed at a very early stage.
- **Operationality:** It is important that each alternative can be judged against each criterion.

Criteria characteristics

- **Mutual independence of criteria:**

Straightforward applications of MCDM require that preferences associated with the consequences of the alternatives are independent of each other from one criterion to the next.

- **Number of criteria:** An excessive number of criteria leads to extra analytical effort in assessing input data and can make communication of the results of the analysis more difficult.

WEIGHTING THE CRITERIA:

- **Direct Determination**
 - Rating, Point allocation, Categorization
 - Ranking
 - Swing
 - Trade-off
 - Ratio (Eigenvector prioritization)
- **Indirect Determination**
 - Centrality
 - Regression – Conjoint analysis
 - Interactive

WEIGHTING THE CRITERIA:

- **The ranking method:** In this method, the criteria are simply ranked in perceived order Of importance by decision- makers: $c_1 > c_2 > c_3 > \dots > c_i$. The method assumes that the weights are non-negative and sum to 1.
- **Rating method:** The point allocation approach is based on allocating points ranging from 0 to 100, where 0 indicates that the criterion can be ignored, and 100 represents the situation where only one criterion need to be considered. In ratio estimation procedure which is a modification of the point allocation method. A score of 100 is assigned to the most important criterion and proportionally smaller weights are given to criteria lower in the order. The score assigned for the least important attribute is used to calculate the ratios.

WEIGHTING THE CRITERIA:

- **Pair wise comparison method:** involves pair wise comparisons to create a ratio matrix. It uses scale table for pair wise comparisons and then computes the weights.

Scale for pairwise comparison (Saaty, 1980)

Intensity of importance	Definition
1	Equal importance
2	Equal to moderately importance
3	Moderate importance
4	Moderate to strong importance
5	Strong importance
6	Strong to very strong importance
7	Very strong importance
8	Very to extremely strong importance
9	Extreme importance

STANDARDIZING THE RAW SCORES

- Because usually the various criteria are measured in different units, the scores in the evaluation matrix S have to be transformed to a normalized scale. some methods are :

- $\alpha_{ij} = S_{ij} / \sum(S_{ij})$

- $\alpha_{ij} = S_{ij} / \max(S_{ij})$

- $\alpha_{ij} = (S_{ij} - \min S_{ij}) / (\max S_{ij} - \min S_{ij})$

- $\alpha_{ij} = S_{ij} / \sqrt{\sum(S_{ij})^2}$

PROBLEM SOLVING TECHNIQUES

Some problem solving techniques are :

- **SAW** (Simple Additive Weighting)
- **TOPSIS** (Technique for Order Preference by Similarity to the Ideal Solution)
- **ELECTRE** (Elimination et Choice Translating Reality)
- **BAYESIAN NETWORK BASED FRAMEWORK**
- **AHP** (The Analytical Hierarchy Process)
- **SMART** (The Simple Multi Attribute Rating Technique)
- **ANP** (Analytic network process)

The selection of the models are based on the following evaluation criteria suggested by Dodgson et al. (2001):

- internal consistency and logical soundness;
- transparency;
- ease of use;
- data requirements are consistent with the importance of the issue being considered;
- realistic time and manpower resource requirements for the analytical process;
- ability to provide an audit trail; and
- software availability, where needed.

SAW (SIMPLE ADDITIVE WEIGHTING):

Multiplies the normalized value of the criteria for the alternatives with the importance of the criteria .the alternative with the highest score is selected as the preferred one.

SAW (SIMPLE ADDITIVE WEIGHTING):

$$S_i = \sum_{j=1}^M w_j r_{ij} \quad \text{for } i = 1, 2, \dots, N$$

where

S_i is the overall score of the i^{th} alternative;

r_{ij} is the normalised rating of the i^{th} alternative for the j^{th} criterion, which is computed as $r_{ij} = x_{ij} / (\max_i x_{ij})$ for the benefit and $r_{ij} = (1/x_{ij}) / [\max_i (1/x_{ij})]$ for the cost criterion representing an element of the normalised matrix $\mathbf{R}$;

x_{ij} is an element of the Decision Matrix $\mathbf{A}$, which represents the original value of the j^{th} criterion of the i^{th} alternative;

w_j is the importance (weight) of the j^{th} criterion;

N is the number of alternatives;

M is the number of criteria.

A SIMPLE EXAMPLE OF USING SAW METHOD

- Objective
 - Selecting a car
- Criteria
 - Style, Reliability, Fuel-economy
- Alternatives
 - Civic Coupe, Saturn Coupe, Ford Escort, Mazda Miata

WEIGHTS AND SCORES

Weight	0.3	0.4	0.3	Si
	Style	Reliability	Fuel Eco.	
Civic	7	9	9	8.4
Saturn	8	7	8	7.6
Ford	9	6	8	7.5
Mazda	6	7	8	7.0

TOPSIS (TECHNIQUE FOR ORDER PREFERENCE BY SIMILARITY TO THE IDEAL SOLUTION)

- In this method two artificial alternatives are hypothesized:
- **Ideal alternative**: the one which has the best level for all attributes considered.
- **Negative ideal alternative**: the one which has the worst attribute values.
- TOPSIS selects the alternative that is the closest to the ideal solution and farthest from negative ideal alternative.

INPUT TO TOPSIS

- TOPSIS assumes that we have m alternatives (options) and n attributes/criteria and we have the score of each option with respect to each criterion.
- Let x_{ij} score of option i with respect to criterion j
We have a matrix $X = (x_{ij})$ $m \times n$ matrix.
- Let J be the set of benefit attributes or criteria (more is better)
- Let J' be the set of negative attributes or criteria (less is better)

STEPS OF TOPSIS

- Step 1: Construct normalized decision matrix.
- This step transforms various attribute dimensions into non-dimensional attributes, which allows comparisons across criteria.
- Normalize scores or data as follows:

$$r_{ij} = x_{ij} / \sqrt{\sum_i x_{ij}^2} \text{ for } i = 1, \dots, m; j = 1, \dots, n$$

STEPS OF TOPSIS

- **Step 2:** Construct the weighted normalized decision matrix.
- Assume we have a set of weights for each criteria w_j for $j = 1, \dots, n$.
- Multiply each column of the normalized decision matrix by its associated weight.
- An element of the new matrix is:

$$v_{ij} = w_j r_{ij}$$

STEPS OF TOPSIS

- Step 3: Determine the ideal and negative ideal solutions.

- Ideal solution.

$A^* = \{ v_1^*, \dots, v_n^* \}$, where

$$v_j^* = \{ \max_i (v_{ij}) \text{ if } j \in J ; \min_i (v_{ij}) \text{ if } j \in J' \}$$

- Negative ideal solution.

$A' = \{ v_1', \dots, v_n' \}$, where

$$v' = \{ \min_i (v_{ij}) \text{ if } j \in J ; \max_i (v_{ij}) \text{ if } j \in J' \}$$

STEPS OF TOPSIS

- **Step 4:** Calculate the separation measures for each alternative.

- The separation from the ideal alternative is:

$$S_i^* = \left[\sum_j (v_j^* - v_{ij})^2 \right]^{1/2} \quad i = 1, \dots, m$$

- Similarly, the separation from the negative ideal alternative is:

$$S'_i = \left[\sum_j (v'_j - v_{ij})^2 \right]^{1/2} \quad i = 1, \dots, m$$

STEPS OF TOPSIS

- **Step 5:** Calculate the relative closeness to the ideal solution C_i^*

$$C_i^* = S'_i / (S_i^* + S'_i) , \quad 0 < C_i^* < 1$$

Select the Alternative with C_i^* closest to 1.

AN EXAMPLE OF USING TOPSIS METHOD

Weight	0.1	0.4	0.3	0.2
	Style	Reliability	Fuel Eco.	Cost
Civic	7	9	9	8
Saturn	8	7	8	7
Ford	9	6	8	9
Mazda	6	7	8	6

STEPS OF TOPSIS

- **Step 1:** calculate $(\sum x_{ij}^2)^{1/2}$ for each column and divide each column by that to get r_{ij}

	Style	Rel.	Fuel	Cost
Civic	0.46	0.61	0.54	0.53
Saturn	0.53	0.48	0.48	0.46
Ford	0.59	0.41	0.48	0.59
Mazda	0.40	0.48	0.48	0.40

STEPS OF TOPSIS

- Step 2 : multiply each column by w_j to get v_{ij} .

	Style+	Rel.+	Fuel+	Cost -
Civic	0.046	0.244	0.162	0.106
Saturn	0.053	0.192	0.144	0.092
Ford	0.059	0.164	0.144	0.118
Mazda	0.040	0.192	0.144	0.080

STEPS OF TOPSIS

Step 3 (a): determine ideal solution A^* .

$$A^* = \{0.059, 0.244, 0.162, 0.080\}$$

	Style	Rel.	Fuel	Cost
Civic	0.046	0.244	0.162	0.106
Saturn	0.053	0.192	0.144	0.092
Ford	0.059	0.164	0.144	0.118
Mazda	0.040	0.192	0.144	0.080
		↑		↓

STEPS OF TOPSIS

Step 3 (b): find negative ideal solution A'.

$$A' = \{0.040, 0.164, 0.144, 0.118\}$$

	Style	Rel.	Fuel	Cost
Civic	0.046	0.244	0.162	0.106
Saturn	0.053	0.192	0.144	0.092
Ford	0.059	0.164	0.144	0.118
Mazda	0.040	0.192	0.144	0.080
		↓		↑

STEPS OF TOPSIS

Step 4 (a): determine separation from ideal solution $A^* = \{0.059, 0.244, 0.162, 0.080\}$

$S_i^* = [\sum (v_j^* - v_{ij})^2]^{1/2}$ for each row j

	Style	Rel.	Fuel	Cost
Civic	$(.046-.059)^2$	$(.244-.244)^2$	$(0)^2$	$(.026)^2$
Saturn	$(.053-.059)^2$	$(.192-.244)^2$	$(-.018)^2$	$(.012)^2$
Ford	$(.053-.059)^2$	$(.164-.244)^2$	$(-.018)^2$	$(.038)^2$
Mazda	$(.053-.059)^2$	$(.192-.244)^2$	$(-.018)^2$	$(.0)^2$

STEPS OF TOPSIS

Step 4 (a): determine separation from ideal solution Si^*

	$\Sigma(v_j^* - v_{ij})^2$	$Si^* = [\Sigma (v_j^* - v_{ij})^2]^{1/2}$
Civic	0.000845	0.029
Saturn	0.003208	0.057
Ford	0.008186	0.090
Mazda	0.003389	0.058

STEPS OF TOPSIS

Step 4: determine separation from negative ideal solution
 S_i'

	$\Sigma(v_j' - v_{ij})^2$	$S_i' = [\Sigma (v_j' - v_{ij})^2]^{1/2}$
Civic	0.006904	0.083
Saturn	0.001629	0.040
Ford	0.000361	0.019
Mazda	0.002228	0.047

STEPS OF TOPSIS

Step 5: Calculate the relative closeness to the ideal solution $C_i^* = S'_i / (S_i^* + S'_i)$

	$S'_i / (S_i^* + S'_i)$	C_i^*		
Civic	0.083/0.112	0.74	←	BEST
Saturn	0.040/0.097	0.41		
Ford	0.019/0.109	0.17		
Mazda	0.047/0.105	0.45		

AHP (THE ANALYTICAL HIERARCHY PROCESS)

- AHP uses a hierarchical structure and pairwise comparisons.
- An AHP hierarchy has at least three levels:
 - 1) the main objective of the problem at the top.
 - 2) multiple criteria that define alternatives in the middle.(m)
 - 3) competing alternatives at the bottom.(n)

STEPS OF AHP

- 1) Criteria weighting must be determined using $(m*(m-1))/2$ pair wise comparisons.
- 2) Alternatives scoring using $m*((n*(n-1))/2)$ pair wise comparisons between alternatives for each criteria.
- 3) After completing pair wise comparisons AHP is just the hierarchical application of SAW.

SCALE OF RELATIVE IMPORTANCE TABLE

Intensity of importance	Definition	Explanation
1	Equal importance	Two activities contribute equally to the objective
3	Weak importance of one over another	Experience and judgement slightly favour one activity over another
5	Essential or strong importance	Experience and judgement strongly favour one activity over another
7	Demonstrated importance	An activity is strongly favoured and its dominance demonstrated in practice
9	Absolute importance	The evidence favouring one activity over another is of the highest possible order of affirmation
2, 4, 6, 8	Intermediate values between the two adjacent judgements	When compromise is needed
Reciprocals of above non-zero	If activity i has one of the above non-zero numbers assigned to it when compared with activity j , then j has the reciprocal value when compared with i	

Scale of relative importances (according to Saaty[6])

ANALYTIC HIERARCHY PROCESS

The Analytic Hierarchy Process (AHP), is a procedure designed to quantify managerial judgments of the relative importance of each of several conflicting criteria used in the decision making process.



ANALYTIC HIERARCHY PROCESS

- Step 1: List the Overall Goal, Criteria, and Decision Alternatives

----- For each criterion, perform steps 2 through 5 -----

- Step 2: Develop a Pairwise Comparison Matrix

Rate the relative importance between each pair of decision alternatives. The matrix lists the alternatives horizontally and vertically and has the numerical ratings comparing the horizontal (first) alternative with the vertical (second) alternative.

Ratings are given as follows:

...



ANALYTIC HIERARCHY PROCESS

○ Step 2: Pairwise Comparison Matrix (continued)

Compared to the second
alternative, the first alternative is: Numerical
rating

extremely preferred	9
very strongly preferred	7
strongly preferred	5
moderately preferred	3
equally preferred	1



ANALYTIC HIERARCHY PROCESS

- Step 2: Pairwise Comparison Matrix (continued)

Intermediate numeric ratings of 8, 6, 4, 2 can be assigned. A reciprocal rating (i.e. $1/9$, $1/8$, etc.) is assigned when the second alternative is preferred to the first. The value of 1 is always assigned when comparing an alternative with itself.



ANALYTIC HIERARCHY PROCESS

- Step 3: Develop a Normalized Matrix

Divide each number in a column of the pairwise comparison matrix by its column sum.

- Step 4: Develop the Priority Vector

Average each row of the normalized matrix. These row averages form the priority vector of alternative preferences with respect to the particular criterion. The values in this vector sum to 1.



ANALYTIC HIERARCHY PROCESS

- Step 5: Calculate a Consistency Ratio

The consistency of the subjective input in the pairwise comparison matrix can be measured by calculating a consistency ratio. A consistency ratio of less than .1 is good. For ratios which are greater than .1, the subjective input should be re-evaluated.

- Step 6: Develop a Priority Matrix

After steps 2 through 5 has been performed for all criteria, the results of step 4 are summarized in a priority matrix by listing the decision alternatives horizontally and the criteria vertically. The column entries are the priority vectors for each criterion.



ANALYTIC HIERARCHY PROCESS

- Step 7: Develop a Criteria Pairwise Development Matrix

This is done in the same manner as that used to construct alternative pairwise comparison matrices by using subjective ratings (step 2). Similarly, normalize the matrix (step 3) and develop a criteria priority vector (step 4).

- Step 8: Develop an Overall Priority Vector

Multiply the criteria priority vector (from step 7) by the priority matrix (from step 6).



DETERMINING THE CONSISTENCY RATIO

- Step 1:

For each row of the pairwise comparison matrix, determine a weighted sum by summing the multiples of the entries by the priority of its corresponding (column) alternative.

- Step 2:

For each row, divide its weighted sum by the priority of its corresponding (row) alternative.

- Step 3:

Determine the average, $\lambda_{\max}$, of the results of step 2.



DETERMINING THE CONSISTENCY RATIO

○ Step 4:

Compute the consistency index, CI, of the n alternatives by: $CI = (\lambda_{\max} - n)/(n - 1)$.

○ Step 5:

Determine the random index, RI, as follows:

<u>Number of Alternative (n)</u>	<u>Random Index (RI)</u>	<u>Number of Alternative (n)</u>	<u>Random Index (RI)</u>
3	0.58	6	1.24
4	0.90	7	1.32
5	1.12	8	1.41

○ Step 6:

Compute the consistency ratio: $CR = CI/RI$.



EXAMPLE: GILL GLASS

Designer Gill Glass must decide which of three manufacturers will develop his "signature" toothbrushes. Three factors seem important to Gill: (1) his costs; (2) reliability of the product; and, (3) delivery time of the orders.

The three manufacturers are Cornell Industries, Brush Pik, and Picobuy. Cornell Industries will sell toothbrushes to Gill Glass for \$100 per gross, Brush Pik for \$80 per gross, and Picobuy for \$144 per gross. Gill has decided that in terms of price, Brush Pik is moderately preferred to Cornell and very strongly preferred to Picobuy. In turn Cornell is strongly to very strongly preferred to Picobuy.



EXAMPLE: GILL GLASS

- Hierarchy for the Manufacturer Selection Problem

Overall Goal

Select the Best Toothbrush Manufacturer

Criteria

Cost

Reliability

Delivery Time

Decision
Alternatives

Cornell
Brush Pik
Picobuy

Cornell
Brush Pik
Picobuy

Cornell
Brush Pik
Picobuy



The accounting department has determined that in terms of criteria, cost is extremely preferable to delivery time and very strongly preferable to reliability, and that reliability is very strongly preferable to delivery time.



- Pairwise Comparison Matrix for Criteria

	Cost	Reliability	Delivery
Cost	1	7	9
Reliability	1/7	1	7
Delivery	1/9	1/7	1



- Normalized Matrix for Criteria

Divide each entry in the pairwise comparison matrix by its corresponding column sum.

	Cost	Reliability	
Delivery			
Cost	$63/79$	$49/57$	$9/17$
Reliability	$9/79$	$7/57$	$7/17$
Delivery	$7/79$	$1/57$	$1/17$



○ Priority Vector For Criteria

The priority vector is determined by averaging the row entries in the normalized matrix. Converting to decimals we get:

$$\begin{array}{lll} \text{Cost:} & (63/79 + 49/57 + 9/17)/3 & = \left[.729 \right] \\ \text{Reliability:} & (9/79 + 7/57 + 7/17)/3 & = \left[.216 \right] \\ \text{Delivery:} & (7/79 + 1/57 + 1/17)/3 & = \left[.055 \right] \end{array}$$



DETERMINE INCONSISTENCY RATIO (CR)

○ Checking Consistency

- Multiply each column of the pairwise comparison matrix by its priority:

$$0.729 \begin{bmatrix} 1 \\ 1/7 \\ 1/9 \end{bmatrix} + .216 \begin{bmatrix} 7 \\ 1 \\ 1/7 \end{bmatrix} + .055 \begin{bmatrix} 9 \\ 7 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.736 \\ 0.705 \\ 0.167 \end{bmatrix}$$

- Divide these number by their priorities to get:

$$2.736/0.729 = 3.753$$

$$0.705/0.216 = 3.264$$

$$0.167/0.055 = 3.034$$



○ Checking Consistency

- Average the above results to get $\lambda_{\max}$.

$$\lambda_{\max} = (3.753 + 3.264 + 3.034)/3 = 3.35$$

- Compute the consistence index, CI, for two terms.

$$CI = (\lambda_{\max} - n)/(n - 1) = (3.35 - 3)/2 = 0.175$$

- Compute the consistency ratio, CR, by CI/RI, where RI = .58 for 3 factors:

$$CR = CI/RI = 0.175/0.58 = 0.30$$

Since the consistency ratio, CR, is > 0.10 , this should not be acceptable range for consistency.



EXAMPLE: GILL GLASS

- Forming the Pairwise Comparison Matrix For Cost
 - Since Brush Pik is moderately preferred to Cornell, Cornell's entry in the Brush Pik row is 3 and Brush Pik's entry in the Cornell row is $1/3$.
 - Since Brush Pik is very strongly preferred to Picobuy, Picobuy's entry in the Brush Pik row is 7 and Brush Pik's entry in the Picobuy row is $1/7$.
 - Since Cornell is strongly to very strongly preferred to Picobuy, Picobuy's entry in the Cornell row is 6 and Cornell's entry in the Picobuy row is $1/6$.



EXAMPLE: GILL GLASS

- Pairwise Comparison Matrix for Cost

	Cornell	Brush Pik	Picobuy
Cornell	1	$1/3$	6
Brush Pik	3	1	7
Picobuy	$1/6$	$1/7$	1



- Normalized Matrix for Cost

Divide each entry in the pairwise comparison matrix by its corresponding column sum. For example, for Cornell the column sum = $1 + 3 + 1/6 = 25/6$. This gives:

	Cornell	Brush Pik	Picobuy
Cornell	$6/25$	$7/31$	$6/14$
Brush Pik	$18/25$	$21/31$	$7/14$
Picobuy	$1/25$	$3/31$	$1/14$



○ Priority Vector For Cost

The priority vector is determined by averaging the row entries in the normalized matrix.

Converting to decimals we get:

$$\begin{array}{lcl} \text{Cornell:} & (6/25 + 7/31 + 6/14)/3 & = \begin{bmatrix} .298 \end{bmatrix} \\ \text{Brush Pik:} & (18/25 + 21/31 + 7/14)/3 & = \begin{bmatrix} .632 \end{bmatrix} \\ \text{Picobuy:} & (1/25 + 3/31 + 1/14)/3 & = \begin{bmatrix} .069 \end{bmatrix} \end{array}$$



○ Checking Consistency

- Multiply each column of the pairwise comparison matrix by its priority:

$$.298 \begin{bmatrix} 1 \\ 3 \\ 1/6 \end{bmatrix} + .632 \begin{bmatrix} 1/3 \\ 1 \\ 1/7 \end{bmatrix} + .069 \begin{bmatrix} 6 \\ 7 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.923 \\ 2.009 \\ 0.209 \end{bmatrix}$$

- Divide these number by their priorities to get:

$$.923/.298 = 3.097$$

$$2.009/.632 = 3.179$$

$$.209/.069 = 3.029$$



○ Checking Consistency

- Average the above results to get $\lambda_{\max}$.

$$\lambda_{\max} = (3.097 + 3.179 + 3.029)/3 = 3.102$$

- Compute the consistence index, CI, for two terms.

$$CI = (\lambda_{\max} - n)/(n - 1) = (3.102 - 3)/2 = .051$$

- Compute the consistency ratio, CR, by CI/RI, where RI = .58 for 3 factors:

$$CR = CI/RI = .051/.58 = .088$$

Since the consistency ratio, CR, is less than .10, this is well within the acceptable range for consistency.



Gill Glass has determined that for reliability, Cornell is very strongly preferable to Brush Pik and equally preferable to Picobuy. Also, Picobuy is strongly preferable to Brush Pik.



- Pairwise Comparison Matrix for Reliability

	Cornell	Brush Pik	Picobuy
Cornell	1	7	2
Brush Pik	1/7	1	5
Picobuy	1/2	1/5	1



○ Normalized Matrix for Reliability

Divide each entry in the pairwise comparison matrix by its corresponding column sum. For example, for Cornell the column sum = $1 + 1/7 + 1/2 = 23/14$. This gives:

	Cornell	Brush Pik	Picobuy
Cornell	$14/23$	$35/41$	$2/8$
Brush Pik	$2/23$	$5/41$	$5/8$
Picobuy	$7/23$	$1/41$	$1/8$



○ Priority Vector For Reliability

The priority vector is determined by averaging the row entries in the normalized matrix. Converting to decimals we get:

$$\text{Cornell: } (14/23 + 35/41 + 2/8)/3 = .571$$

$$\text{Brush Pik: } (2/23 + 5/41 + 5/8)/3 = .278$$

$$\text{Picobuy: } (7/23 + 1/41 + 1/8)/3 = .151$$

○ Checking Consistency

Gill Glass' responses to reliability could be checked for consistency in the same manner as was cost.



Gill Glass has determined that for delivery time, Cornell is equally preferable to Picobuy. Both Cornell and Picobuy are very strongly to extremely preferable to Brush Pik.



- Pairwise Comparison Matrix for Delivery Time

	Cornell	Brush Pik	Picobuy
Cornell	1	8	1
Brush Pik	1/8	1	1/8
Picobuy	1	8	1



- Normalized Matrix for Delivery Time

Divide each entry in the pairwise comparison matrix by its corresponding column sum.

	Cornell	Brush Pik	Picobuy
Cornell	$8/17$	$8/17$	$8/17$
Brush Pik	$1/17$	$1/17$	$1/17$
Picobuy	$8/17$	$8/17$	$8/17$



○ Priority Vector For Delivery Time

The priority vector is determined by averaging the row entries in the normalized matrix. Converting to decimals we get:

$$\text{Cornell: } (8/17 + 8/17 + 8/17)/3 = .471$$

$$\text{Brush Pik: } (1/17 + 1/17 + 1/17)/3 = .059$$

$$\text{Picobuy: } (8/17 + 8/17 + 8/17)/3 = .471$$

○ Checking Consistency

Gill Glass' responses to delivery time could be checked for consistency in the same manner as was cost.



The accounting department has determined that in terms of criteria, cost is extremely preferable to delivery time and very strongly preferable to reliability, and that reliability is very strongly preferable to delivery time.



- Pairwise Comparison Matrix for Criteria

	Cost	Reliability	Delivery
Cost	1	7	9
Reliability	1/7	1	7
Delivery	1/9	1/7	1



- Normalized Matrix for Criteria

Divide each entry in the pairwise comparison matrix by its corresponding column sum.

	Cost	Reliability	
Delivery			
Cost	$63/79$	$49/57$	$9/17$
Reliability	$9/79$	$7/57$	$7/17$
Delivery	$7/79$	$1/57$	$1/17$



Overall Priority Vector

The overall priorities are determined by multiplying the priority vector of the criteria by the priorities for each decision alternative for each objective.

Priority Vector

for Criteria [.729 .216 .055]

	Cost	Reliability	Delivery
Cornell	.298	.571	.471
Brush Pik	.632	.278	.059
Picobuy	.069	.151	.471



- Overall Priority Vector (continued)

Thus, the overall priority vector is:

$$\begin{array}{lcl} \text{Cornell:} & (.729)(.298) + (.216)(.571) + (.055)(.471) = & \left[\begin{array}{c} .366 \\ .524 \\ .109 \end{array} \right] \\ \text{Brush Pik:} & (.729)(.632) + (.216)(.278) + (.055)(.059) = & \\ \text{Picobuy:} & (.729)(.069) + (.216)(.151) + (.055)(.471) = & \end{array}$$

Brush Pik appears to be the overall recommendation.



SOME AHP METHOD SHORTCOMINGS

- **Comparison inconsistencies:**
decision-makers using AHP often make inconsistent pair wise comparisons.
- **Rank reversals**
changing of relative alternative rankings due to the addition and deletion of alternatives.
- **Large number of comparisons**
where there are either a large number of attributes and/or alternatives to be evaluated.

SMART (THE SIMPLE MULTI ATTRIBUTE RATING TECHNIQUE)

- In a general sense, SMART is somewhat like AHP in that a hierarchical structure is created to assist in defining a problem and to organize criteria. However, there are some significant differences between two techniques:
 - 1) SMART uses a different terminology. For example, in SMART the lowest level of criteria in the value tree (or objective hierarchy) are called **attributes** rather than sub-criteria and the values of the standardized scores assigned to the attributes derived from value functions are called **ratings**.

- 2) The difference between a value tree in SMART and a hierarchy in AHP is that the value tree has a true tree structure, allowing one attribute or sub-criterion to be connected to only one higher level criterion.
- 3) SMART does not use a relative method for standardizing raw scores to a normalized scale. Instead, a **value function** explicitly defines how each value is transformed to the common model scale. The value function mathematically transforms ratings into a consistent internal scale with lower limit 0, and upper limit 1.

ELECTRE METHOD

- **ELECTRE** developed by Roy (1965) stands for ELimination Et Choix Traduisant la REalit'e (ELimination and Choice Expressing the REality),
- ELECTRE is a kind of outranking method.
- ELECTRE based on two indexes: concordance and discordance.
- If consider two alternatives ***a*** and ***b***, *outranking* defined as follows:
 - ***aSb*** and ***not bSa*** $\leftrightarrow$ ***a P b*** (*a is strictly preferred to b*).
 - ***bSa*** and ***not aSb*** $\leftrightarrow$ ***b P a*** (*b is strictly preferred to a*).
 - ***aSb*** and ***bSa*** $\leftrightarrow$ ***a I b*** (*a is indifferent to b*).
 - ***Not aSb*** and ***not bSa*** $\leftrightarrow$ ***a R b*** (*a is incomparable to b*).
- If we define Concordance index $C(a,b)$ and Discordance index is $D(a,b)$ then:
 - $C(a,b) > C^*$ and $D(a,b) < D^*$ $\leftrightarrow$ ***a P b***

INTRODUCTION TO ELECTRE

- July 1966, Roy presented a paper in Rome in which he used his formal training in Mathematics to develop a practical decision making system, now known as ELECTRE (Elimination Et Choice Translating REality)
- 6 main ELECTRE models : I, II, III, IV, Tri and IS

ELECTRE I

The first version of model devised by Roy (1968):

- Solve multi-criteria problems.
- Choose a set of preference alternatives.

ELECTRE I (CONT)

Include two stages:

- **Construction of the outranking relation :**
aims at com-paring in a comprehensive way each pair of actions.
- **Exploitation of the ranking relationship:**
used to elaborate (explain) recommendations from the results obtained in the first phase. The nature of the recommendations depends on the problematic (choosing, ranking or sorting).

STAGE 1: CONSTRUCTION OF THE OUTRANKING RELATION

Concordance Index :

$$C(a, b) = \frac{\sum_{i \in Q(a, b)} w_i}{\sum_{i=1}^m w_i}$$

$C(a, b) \in [0, 1]$.

$C(a, b)$: measure how much **a** is at least as good as **b**.

$g_i(k)$: the preferred scores of the i^{th} criterion for the k^{th} alternative.

$Q(a, b)$: the set of criteria for which a is equal or preferred to b .

W_i : weight of criterion i^{th}

$$\text{Discordance Index: } D(a, b) = \frac{\max_{i \in R(a, b)} [w_i (g_i(b) - g_i(a))]}{\max_{c, d \in A} [w_i (g_i(c) - g_i(d))]}$$

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$D(a, b) \in [0, 1]$.

$D(a, b)$: measure the degree to which **b** is strictly preferred to **a**

$g_i(k)$: the preferred scores of the i^{th} attribute for the k^{th} alternative

$R(a, b)$: the set of criteria for which b is strictly preferred to a .

A : the set of all alternatives.

OUTRANKING RELATION

If $C(a,b) \geq C^*$ and $D(a,b) \leq D^*$
then a outrank than b , otherwise a does not outrank b .

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Where:

C^* : the concordance thershold

D^* : the discordance thershold

C^*, D^* : defined by DM

$0 < D^*, C^* < 1$

OUTRANK RELATION (CONT)

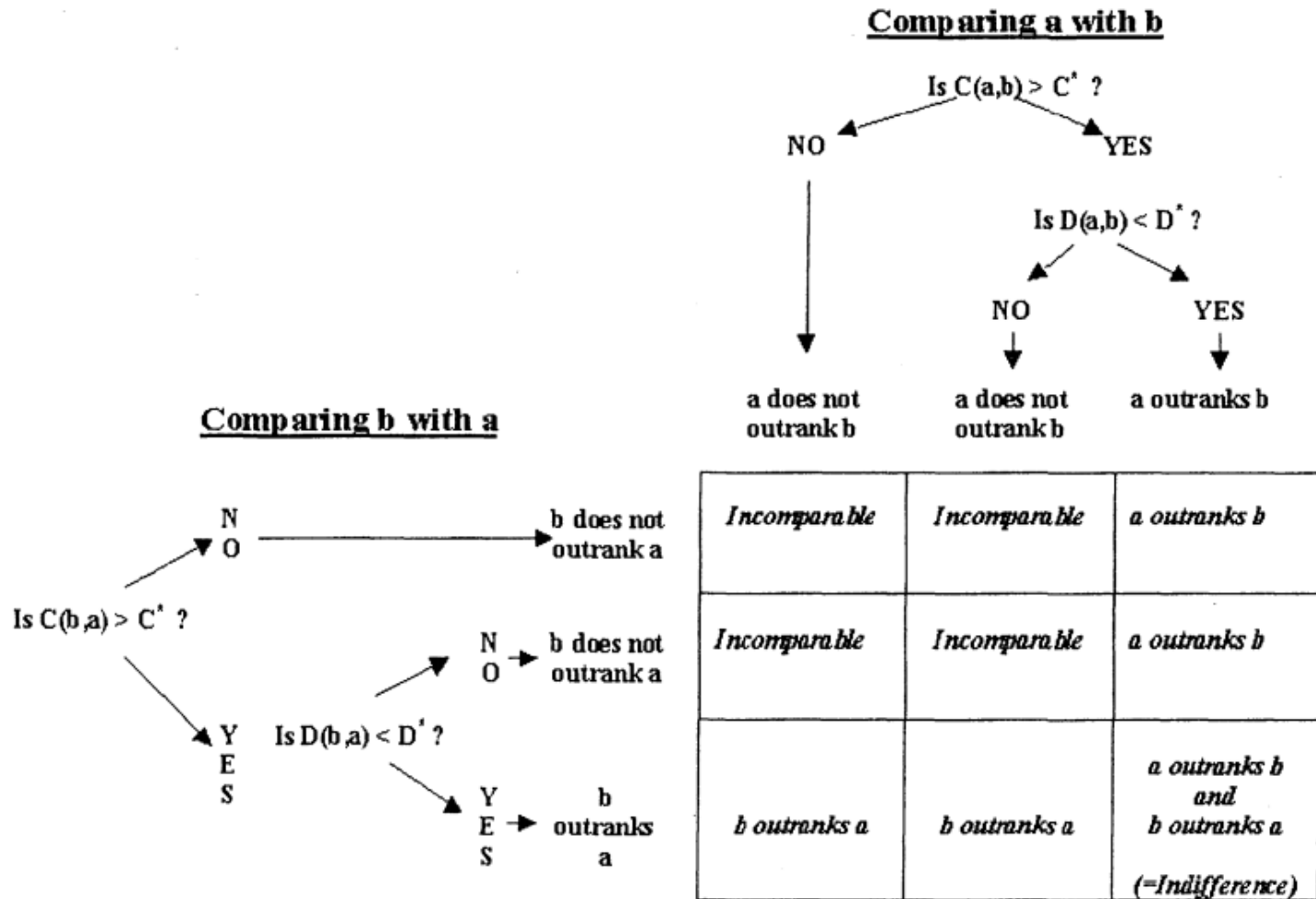
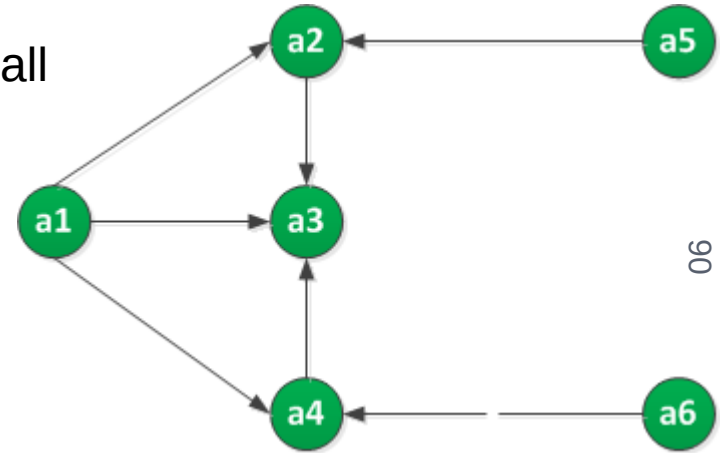


Fig I -1. Building an outranking relation

Stage 2: Exploitation of the Ranking Relationships

Electre I, seeks to obtain a partition of the set of all options (A) into two subsets N and A\N:

- Each option in A\N is outranked by at least one option in N
- The options in N are incomparable



$$A \setminus N \subset A$$

$$N \subset A$$

$$\begin{cases} \forall b \in A \setminus N, \exists a \in N : a S b \\ \forall a, b \in N, a \not S b, b \not S a. \end{cases}$$

Fig 1. Building and exploiting the outranking relation

$$A = \{a_1, a_2, a_3, a_4, a_5, a_6\}$$

$$A \setminus N = \{a_1, a_2, a_4\}$$

$$N = \{a_3, a_5, a_6\}$$

N: a kernel of the graph (the set of preferred alternatives)

RANKING PROCEDURE OF ELECTRE I

- Step 1: Collect the weight of criteria w_i given by DM

	C1	C2	C3
Weighting	3	2	1

- Step 2: Collect the preferred score $g_j(k)$ of the j^{th} criterion for k^{th} alternatives

$g_i(k)$	c1	c2	c3
a	7	10	4
b	5	6	8
c	7	6	11

a is equal or preferred to b

- Step 3: Determine Concordance Index

$$C(a,b) = \frac{\sum_{i \in Q(a,b)} w_i}{\sum_{i=1}^m w_i}$$

$$C(a,b) = \frac{3+2}{3+2+1} = 0.833$$

$$C(b,a) = \frac{1}{3+2+1} = 0.166$$

$$C(a,c) = \frac{3+2}{3+2+1} = 0.833$$

$$C(c,a) = \frac{1}{3+2+1} = 0.166$$

$$C(b,c) = \frac{2}{3+2+1} = 0.333$$

$$C(c,b) = \frac{2+1}{3+2+1} = 0.5$$

C	a	b	c
a	X	0.833	0.833
b	0.166	X	0.333
c	0.166	0.5	X

- Step 4: Determine Discordance Index

$$D(a,b) = \frac{\max_{i \in R(a,b)} [w_i(g_i(b) - g_i(a))]}{\max_{c,d \in A} [w_i(g_i(c) - g_i(d))]}$$

D	a	b	c
a	X	0.5	0.875
b	1	X	0.75
c	1	0	X

$$D(a,b) = \frac{1 \times (8-4)}{\underbrace{2 \times (10-6)}_{\text{The largest range}}} = 0.5$$

$$D(b,a) = \frac{2 \times (10-6)}{2 \times (10-6)} = 1$$

$$D(b,c) = \frac{3 \times (7-5)}{2 \times (10-6)} = 0.75$$

$$D(c,b) = \frac{0}{2 \times (10-6)} = 0$$

$$D(a,c) = \frac{1 \times (11-4)}{2 \times (10-6)} = 0.875$$

$$D(c,a) = \frac{2 \times (10-6)}{2 \times (10-6)} = 1$$

- Step 5: DMs define concordance threshold and discordance threshold

$$C^*; D^*$$

- Step 6: Check condition
- Step 7: Construct the outranking relation graph
- Step 8: Determine the kernel of the graph

ELECTRE II

- ELECTRE II system (Roy and Bertier, 1971, 1973) differs from ELECTRE I
- Rank the options from best to worst, rather than simply establishing an initial kernel.
- Give a complete ordering of non-dominated options, unlike ELECTRE I which provides a partial ordering of the non-dominated set.

STAGE 1: BUILDING THE OUTRANKING RELATIONS

- ❑ Three concordance thresholds and two discordance thresholds are defined
- ❑ A strong outranking relation S^F and a weak outranking relation S^f are built

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1. Strong outranking a S^F b

$$\begin{aligned} C(a, b) &\geq c^+ \\ g_j(a) - g_j(b) &\leq d_1 \\ \frac{P^+(a, b)}{P^-(a, b)} &\geq 1 \end{aligned}$$

c^+, c^0, c^- : three decreasing level of concordance : $0 \leq c^- \leq c^0 \leq c^+ \leq 1$

d_1, d_2 : two decreasing levels of discordance : $0 \leq d_2 \leq d_1 \leq 1$

P^+ : the sum of the weights of the criteria on which option a is preferred to option b

P^- : the sum of the weights of the criteria on which option b is preferred to option a

Sum
of

Outranking relations (cont)

And/Or

$$c(a, b) \geq c^0$$

$$g_j(a) - g_j(b) \leq d_2 \quad \forall j$$

$$\frac{P^+(a, b)}{P^+(a, b)} \geq 1$$

Outranking relations (cont)

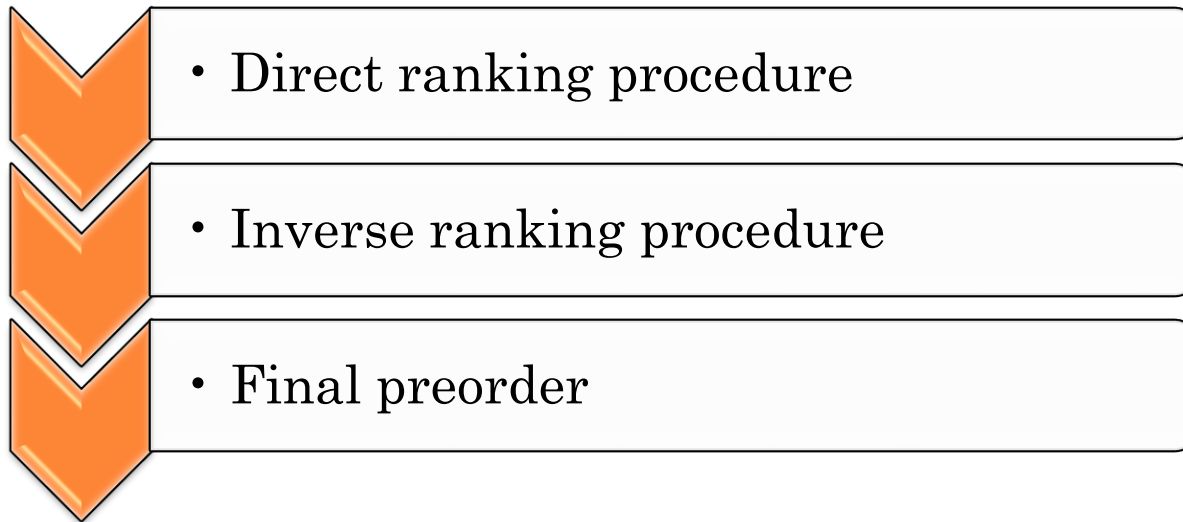
2. Weak outranking: $aS^f b$

$$c(a, b) \geq c^-$$

$$g_j(a) - g_j(b) \leq d_1 \quad \forall j$$

$$\frac{P^+(a, b)}{P^+(a, b)} \geq 1$$

RANKING PROCEDURE OF ELECTRE II



RANKING PROCEDURE OF ELECTRE II

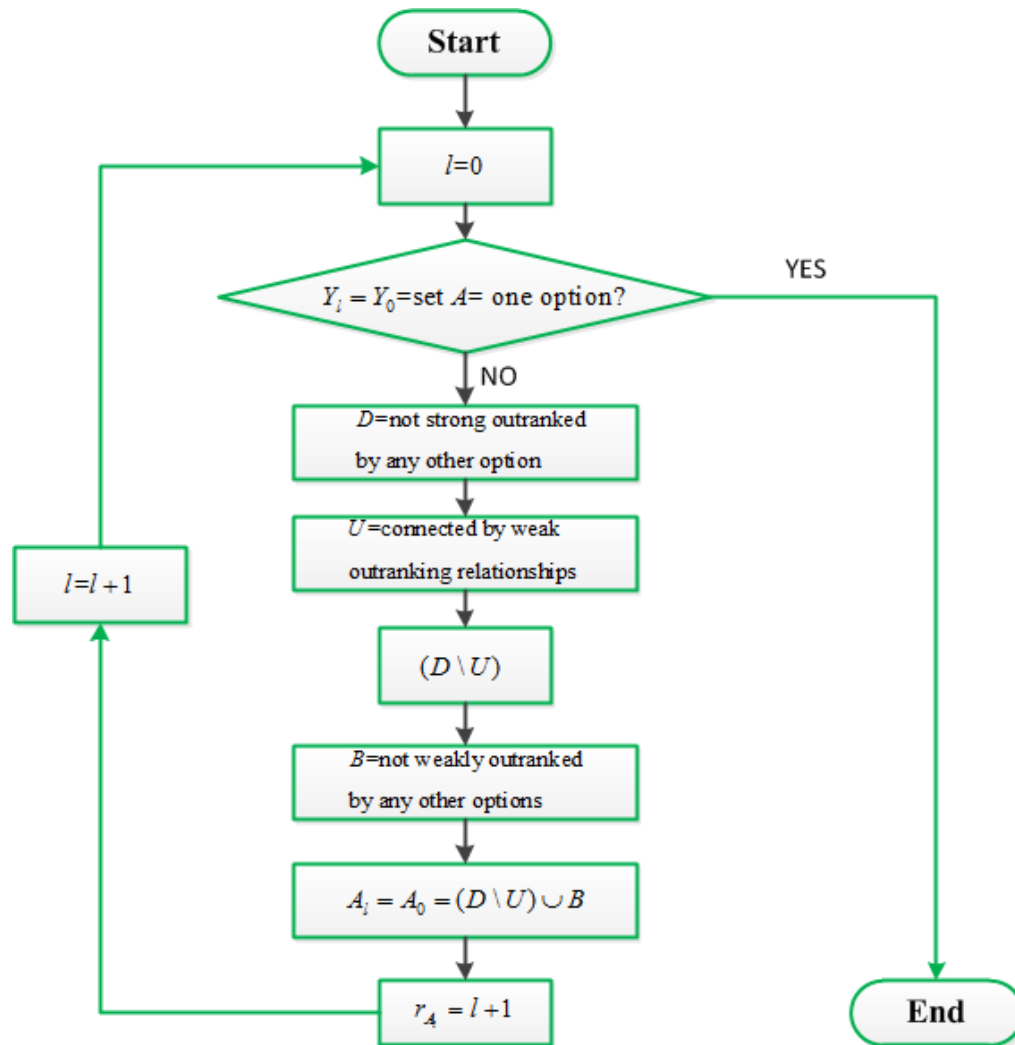


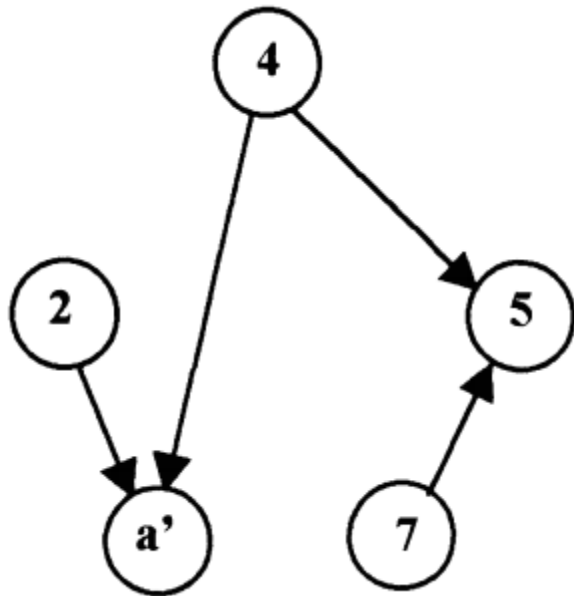
Fig II-1. Flow chart of direct ranking procedure

EXAMPLE FOR ELECTRE II

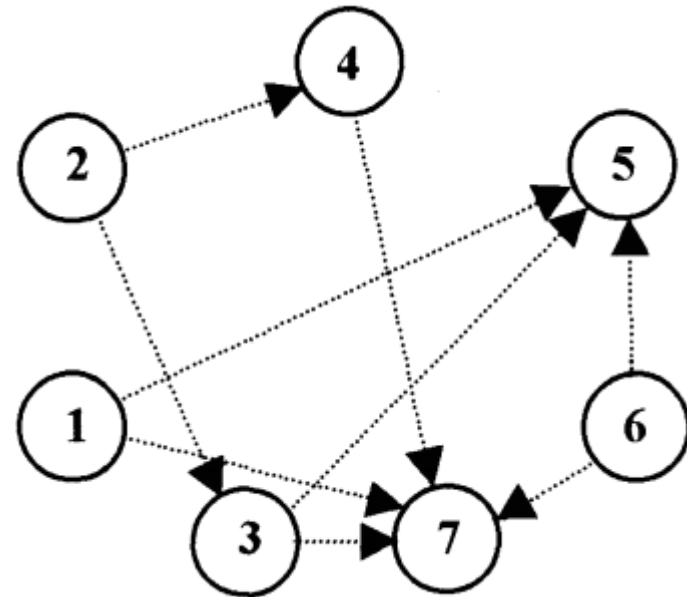
	1	2	3	4	5	6	7
1	---		S^F		S^f	S^F	S^f
2	S^F	---	S^f	S^f		S^F	
3			---		S^f	S^F	S^f
4	S^F		S^F	---	S^F	S^F	S^f
5					---		
6	S^F				S^f	---	S^f
7					S^F		---

Fig II-2. Strong and Weak Outranking Relationships

Source: (M. Rogers et al., ELECTRE and Decision Support
© Springer Science+Business Media New York 2000)



a) Strong outrankings



b) Weak outrankings

Denoted: a' form a circuit (1,3,6)

Fig II-3. The strong and weak outrankings

Direct ranking procedure

○ Step 0:

($I = 0$)

Set A contains seven options, numbered 1 to 7.

$Y_1 = Y_0 =$ all seven options

$D = \{2, 4, 7\}$.

(Options 2, 4 and 7 are not strongly outranked by any of the other option)

$U = \{2, 4, 7\}$.

(All these three options are connected by weak outranking relationships: $2Sf4$ and $4Sf7$)

- $D \setminus U = \emptyset$ (Set D minus set U therefore contains no options)
- $B = \{2\}$ (Option 2 is not weakly outranked by any other option).
- $(D - U) \cup (B) = \{2\}$ Option 2, which is given the rank 1 (1 + 1), and takes no further part in the direct ranking procedure.
- $A_o = \text{Option 2.}$

○ Step 2:

$(I = 1)$

$Y_1 = Y_2 = \{1, 3, 4, 5, 6, 7\}$

$D = \{4, 7\}$.

$U = \{4, 7\}$.

$D \setminus U = \emptyset$

$B = \{4\}$

$(D - U) \cup (B) = \{4\}$ Option 4, which is given the rank 2 ($1 + 1$), and takes no further part in the direct ranking procedure.

$A_1 =$ Option 4. (The best options within step 1)

Step l	Y_l	D	U	B	A_l	$r_{l(l+1)}$
0	1, 2, 3, 4, 5, 6, 7	2, 4, 7	2, 4, 7	2	2	1
1	1, 3, 4, 5, 6, 7	4, 7	4, 7	4	4	2
2	1, 3, 5, 6, 7	1, 3, 6, 7	1, 3, 6, 7	1, 3, 6	1, 3, 6	3
3	5, 7	7	--	--	7	4
4	5	--	--	--	5	5

Tab II-1. Summary of Direct Ranking Procedure

Inverse Ranking Procedure

	1	2	3	4	5	6	7
1	---	S^F		S^F		S^F	
2		---					
3	S^F	S^f	---	S^F			
4		S^f		---			
5	S^f		S^f	S^F	---	S^f	S^F
6	S^F	S^F	S^F	S^F		---	
7	S^f		S^f	S^f		S^f	---

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Tab II-2. Inverted Strong and Weak Outranking Relationships

Step l	Y_l	D	U	B	A_l	$r'_{2(l+1)}$
0	1, 2, 3, 4, 5, 6, 7	1,3,5,6	1,3,5,6	5	5	1
1	1, 2, 3, 4, 6, 7	1,3,6,7	1,3,6,7	7	7	2
2	1, 2, 3, 4, 6	1,3,6	--	--	1,3,6	3
3	2,4	2,4	2,4	4	4	4
4	2	--	--	--	2	5

Tab II-3. Unadjusted Inverse Rankings

$r'_{\max}$

art
of

The adjusted inverse rankings $r_{2(l+1)}$ are estimated:

$$r_{2(l+1)} = 1 + r'_{\max} - r'_{2(l+1)}$$

A_l	$r'_{2(l+1)}$	$r_{2(l+1)}$
5	1	5
7	2	4
1,3,6	3	3
4	4	2
2	5	1

Tab II-4. Adjusted Rankings

$r'_{\max}$: the number of ranking positions in the inverse ranking procedure, equals 5

FINAL PREORDER

The Ranking:

$$2 \rightarrow 4 \rightarrow 1, 3, 6 \rightarrow 7 \rightarrow 5$$

ELECTRE III (Roy, 1978)

- Uses the pseudo-criterion
- Imprecision/uncertainty in the data

OUTRANKING RELATION

Concordance Index $C(a, b) = \frac{1}{W} \sum_{j=1}^n w_j c_j(a, b)$

$$W = \sum_{j=1}^n w_j$$

$$c_j(a, b) = \begin{cases} 1 & \text{if } g_j(a) + q_j(g_j(a)) \geq g_j(b) \\ 0 & \text{if } g_j(a) + p_j(g_j(a)) < g_j(b) \\ \frac{g_j(a) - g_j(b) + p_j(g_j(a))}{p_j(q_j(a)) - q_j(q_j(a))}, & \text{otherwise} \end{cases}$$

p_j : strict preference threshold for criterion j

q_j : indifference threshold for criterion j

The definition of discordance uses a veto threshold:

$$g_j(b) \geq g_j(a) + v_j(g_j(a))$$

Discordance Index

$$D_j(a, b) = \begin{cases} 1 & \text{if } g_j(b) > g_j(a) + v_j(g_j(a)) \\ 0 & \text{if } g_j(b) \leq g_j(a) + p_j(g_j(a)) \\ \frac{g_j(b) - g_j(a) - p_j(g_j(a))}{v_j(g_j(a)) - p_j(g_j(a))}, & \text{otherwise} \end{cases}$$

The degree of credibility of outranking of b by a :

$$S(a,b) = \begin{cases} C(a,b) & \text{if } D_j(a,b) \leq C(a,b), \quad \forall j \\ C(a,b) \prod_{j \in J(a,b)} \frac{(1 - D_j(a,b))}{1 - C(a,b)} & , \text{ otherwise} \end{cases}$$
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$J(a,b)$ is the set of criteria for which $D_j(a,b) > C(a,b)$

Exploiting the Outranking Relationships

a outranks b at the cut off level λ_k

aSb iff $S(a,b) > \lambda_k$ and $S(a,b) - S(b,a) > s(S(a,b))$

$\bar{D}_{k+1}$: the subset of options within

A having the greatest qualification score

$s(\lambda) = 0.3 - 0.15\lambda$: the discrimination threshold

$\lambda_{k+1} = \lambda_k - s(\lambda_k)$: the cut off level $k + 1$

QUALIFICATION

The strength of the option $p_A^{\lambda_1}(a)$:

$$p_A^{\lambda_1}(a) = \left| \left\{ b \in A / a S_A^{\lambda_1} b \right\} \right|$$

The weakness of the option $f_A^{\lambda_1}(a)$:

$$f_A^{\lambda_1}(a) = \left| \left\{ b \in A / b S_A^{\lambda_1} a \right\} \right|$$

The qualification of option a in relation to the set of options A , $q_A^{\lambda_1}(a)$:

$$q_A^{\lambda_1}(a) = p_A^{\lambda_1}(a) - f_A^{\lambda_1}(a)$$

RANKING PROCEDURE OF ELECTRE III



- Descending distillation chain



- Ascending order chain



- Final ranking

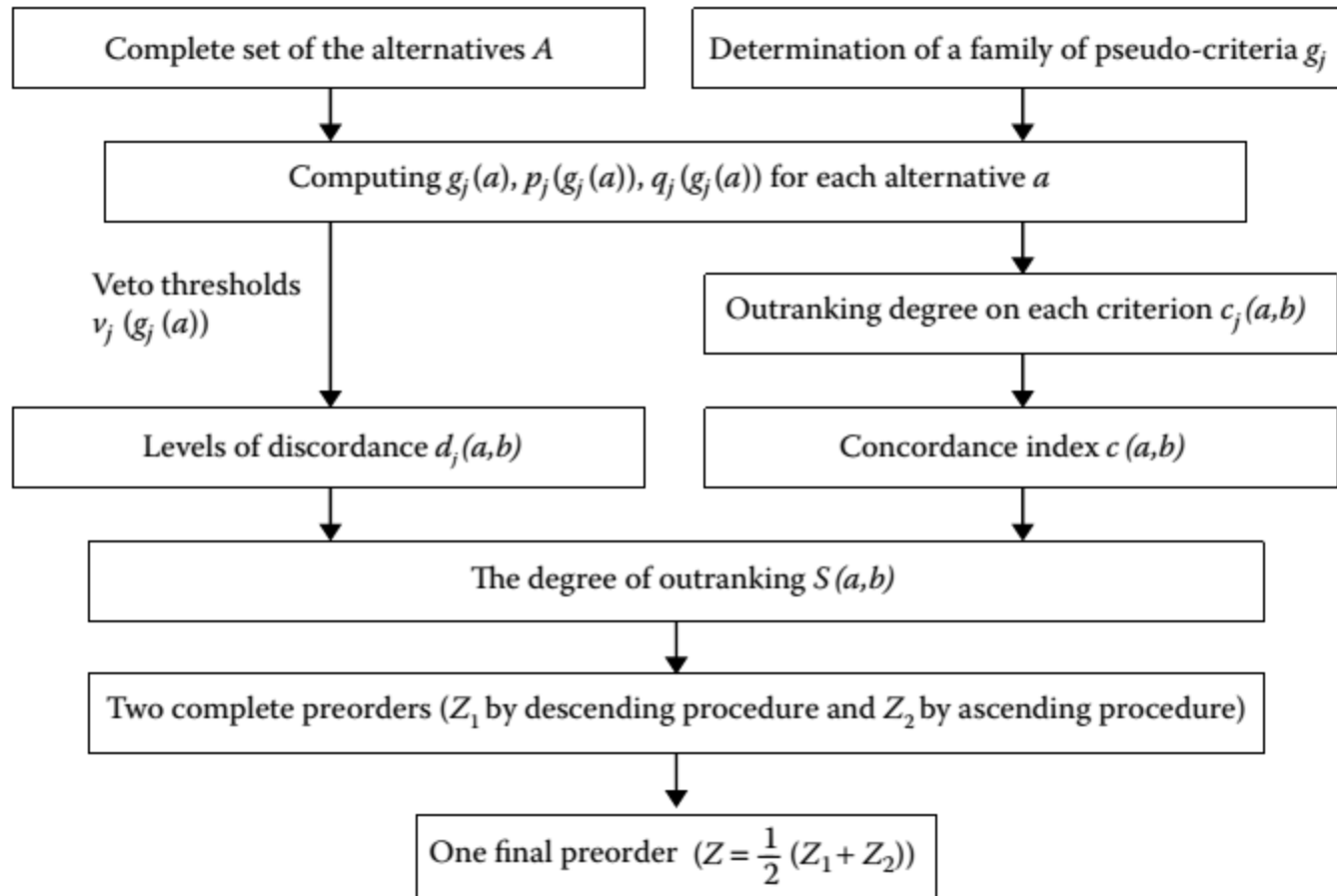


Fig III-1. Procedure of Electre III

(source: Multiple attribute decision making: Methods and Applications)

DESCENDING DISTILLATION CHAIN

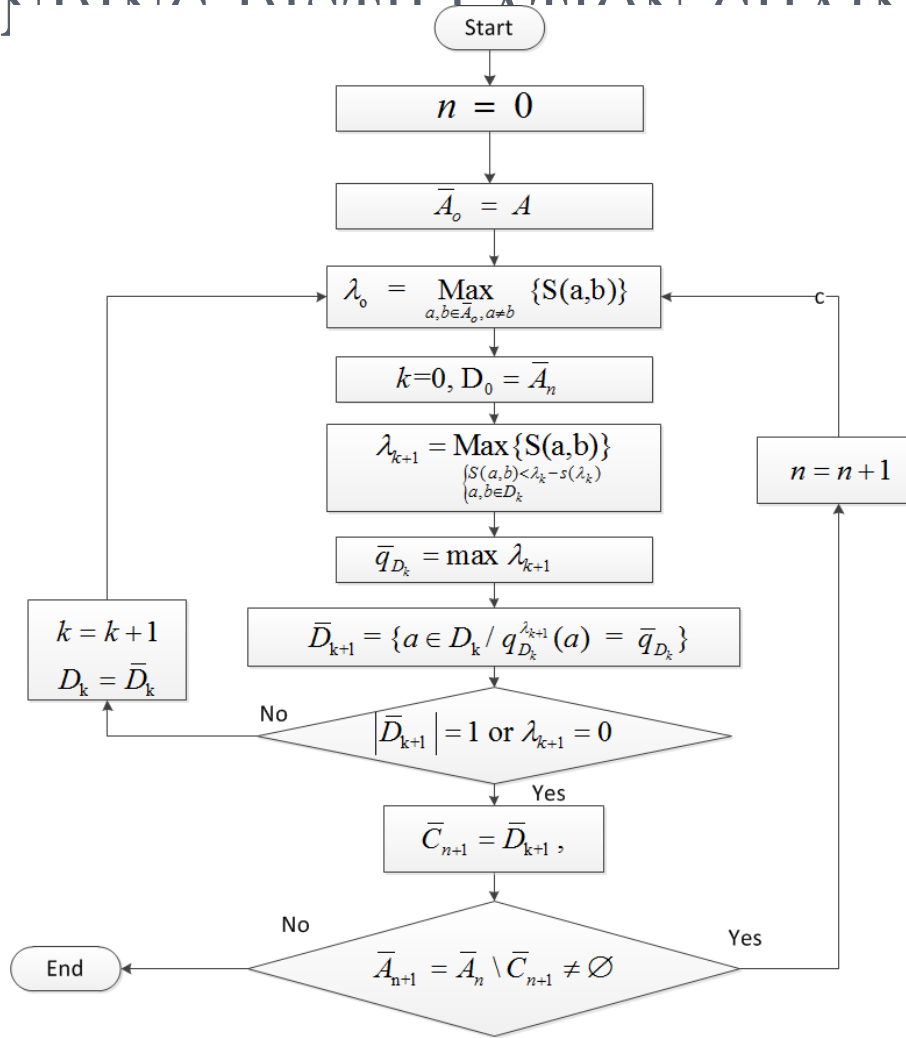


Fig III-2. Flow chart of descending distillation chain

ASCENI

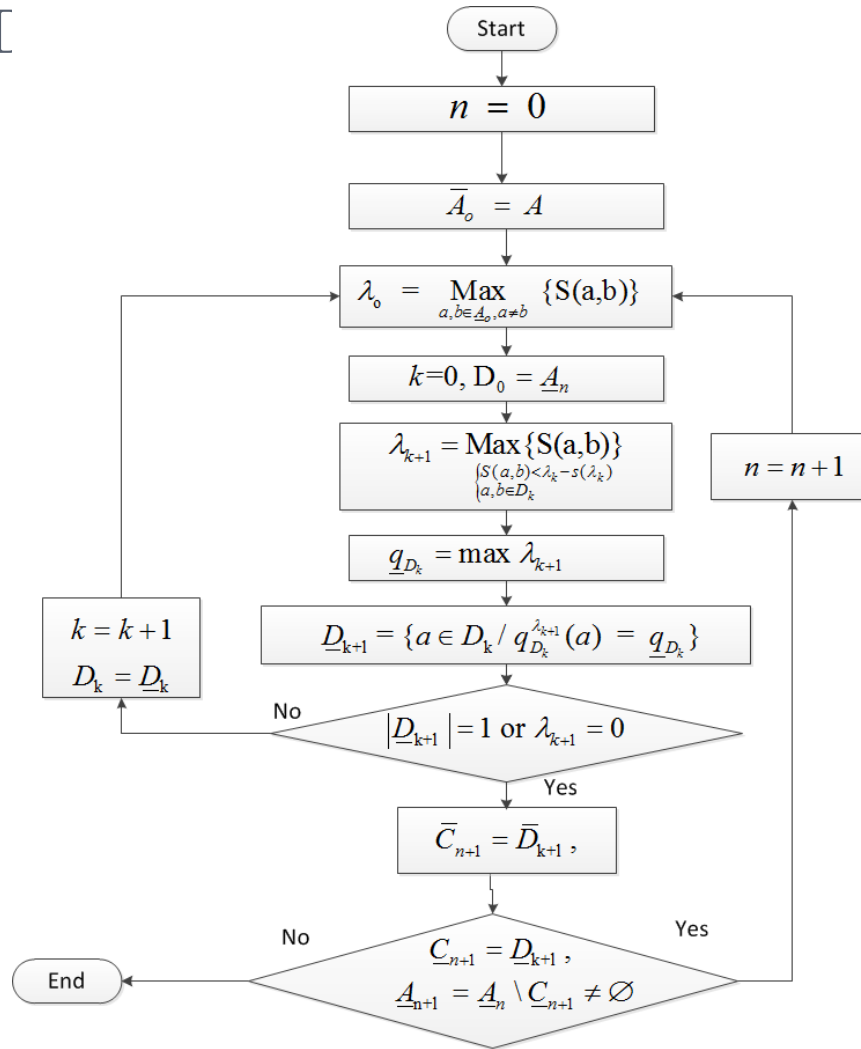


Fig III-3. Flow chart of ascending distillation chain

EXAMPLE FOR ELECTRE III

Descending distillation chain

	S1.1	S1.2	S2.1	S2.2	S2.3	S2.4	S3.1	S3.2	S4.1	S4.2	S4.3
S1.1	1	0.74	0.71	0.71	0.71	0.71	0.74	0.74	0.71	0.68	0.71
S1.2	0.44	1	0.57	0.58	0.58	0.71	0.48	0.57	0.39	0.39	0.71
S2.1	0.36	0.58	1	0.84	0.96	1	0.55	0.69	0.55	0.69	0.83
S2.2	0.36	0.58	1	1	0.95	0.95	0.49	0.65	0.55	0.62	0.76
S2.3	0.38	0.63	0.86	0.68	1	1	0.54	0.68	0.54	0.52	0.68
S2.4	0.38	0.48	0.48	0.47	0.68	1	0.52	0.52	0.52	0.52	0.52
S3.1	0.72	0.86	0.76	0.77	0.75	0.74	1	0.88	0.87	0.84	0.848
S3.2	0.38	0.84	0.74	0.74	0.7	0.87	0.58	1	0.58	0.61	0.87
S4.1	0.35	0.78	0.85	0.85	0.74	0.73	0.67	0.97	1	0.94	0.89
S4.2	0.4	0.81	0.72	0.74	0.81	0.84	0.6	0.98	0.61	1	0.98
S4.3	0.41	0.7	0.74	0.74	0.74	0.87	0.53	0.81	0.55	0.68	1

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$$\lambda_0 = 1$$

Tab III-1. Degree of Credibility Matrix

$$s(\lambda) = 0.3 - 0.15\lambda$$

$$A_0 = A, s(\lambda) = 0.3 - 0.15\lambda$$

• 1st Distillation

Step 1

$$l = 0, D_0 = A_0; \lambda_0 = 1; \lambda_0 - s(\lambda_0) = 0.85; \lambda_1 = 0.848.$$

$$S(2.2, 2.1) = 1 > \lambda_1 = 0.848$$

$$S(2.1, 2.12) = 0.84$$

$$s(S(2.2, 2.1)) = 0.3 - 0.15 * 1 = 0.15$$

$$S(2.2, 2.1) - S(2.1, 2.12) = 1 - 0.84 > 0.15 = s(S(2.2, 2.1))$$

	S1.1	S1.2	S2.1	S2.2	S2.3	S2.4	S3.1	S3.2	S4.1	S4.2	S4.3
S	-	-	2.4	S2.1 S2.3 S2.4	S2.4	-	S1.2 S3.2 S4.1	S2.4	S2.1 S2.2 S3.2 S4.2 S4.3	S3.2 S4.3	S2.4
Strength	0	0	1	3	1	0	3	1	5	2	1
Weakn's	0	1	2	1	1	5	0	3	1	1	2
Qualif'n	0	-1	-1	2	0	-5	3	-2	4	1	-1

$$C_1 = D_1 = \{S4.1\} = 1 \longrightarrow \text{stop}$$

$$A_1 = A_0 \setminus C_1 = \{S1.1, S1.2, S2.1, S2.2, S2.3, S2.4, S3.1, S3.2, S4.2, S4.3\} \neq \emptyset \longrightarrow \text{stop}$$

- 2nd Distillation

Step 1

$l = 0, D_0 = A_1; \lambda_0 = 1; \lambda_0 - s(\lambda_0) = 0.85; \lambda_1 = 0.848$.

	S1.1	S1.2	S2.1	S2.2	S2.3	S2.4	S3.1	S3.2	S4.2	S4.3
S	-	-	2.4	S2.1 S2.3 S2.4	S2.4	-	S1.2 S3.2	S2.4	S3.2 S4.3	S2.4
Strength	0	0	1	3	1	0	2	1	2	1
Weakn's	0	1	1	0	1	5	0	2	0	1
Qualif'n	0	-1	0	3	0	-5	2	-1	2	0

$$C_2 = D_1 = \{S2.2\}$$

$$A_2 = A_1 \setminus C_2 = \{S1.1, S1.2, S2.1, S2.3, S2.4, S3.1, S3.2, S4.2, S4.3\}$$

- 3rd Distillation

Step 1

$l = 0, D_0 = A_2; \lambda_0 = 1; \lambda_0 - s(\lambda_0) = 0.85; \lambda_1 = 0.848 .$

	S1.1	S1.2	S2.1	S2.3	S2.4	S3.1	S3.2	S4.2	S4.3
S	-	-	2.4	S2.4	-	S1.2	S2.4	S3.2	2.4
Strength	0	0	1	1	0	S3.2 2	1	S4.3 2	1
Weakn's	0	1	0	0	4	0	2	0	1
Qualif'n	0	-1	1	1	-4	2	-1	2	0

$D_1 = \{S3.1, S4.2\} = 2 \longrightarrow \text{go to step 2}$

Does S3.1 outrank S4.2 ?

Step 2

$l = 1, D_1 = \{S3.1, S4.2\}; \lambda_1 = 0.848; \lambda_1 - s(\lambda_1) = 0.675; \lambda_2 = 0.53 .$

	S3.1	S4.3
S Strength	S4.3 1	- 0
Weakn's	0	1
Qualif'n	1	-1

$S(S3.1, S4.3) = 0.848$

$S(S4.3, S3.1) = 0.53$

$s(S(S3.1, S4.3)) = (0.3 - (0.15 * 0.848)) = 0.173$

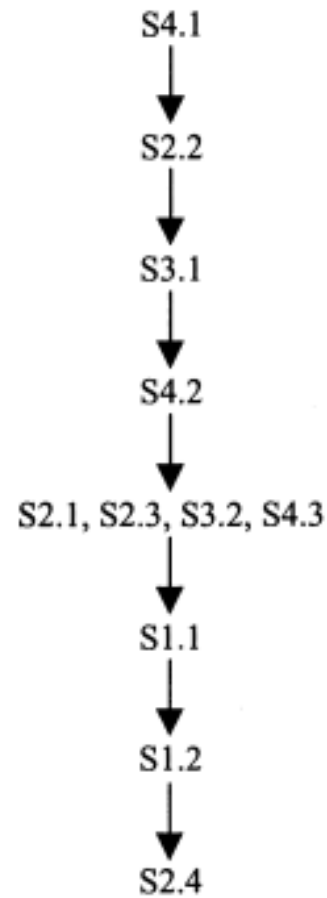
Since $S(S3.1, S4.3) > \lambda_2$, $S(S3.1, S4.3) - S(S4.3, S3.1) > s(S(S3.1, S4.3))$,
S3.1 outranks S4.3

Therefore,

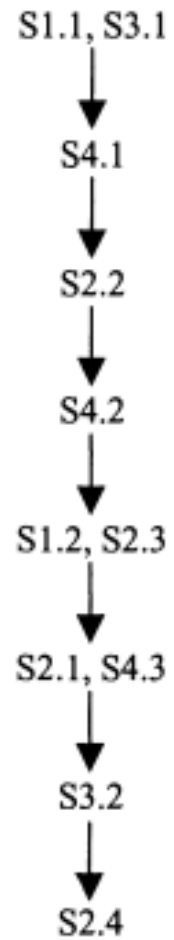
$D_2 = C_3 = \{S3.1\}$

$A_3 = A_2 \setminus C_3 = \{S1.1, S1.2, S2.1, S2.3, S2.4, S3.2, S4.2, S4.3\}$

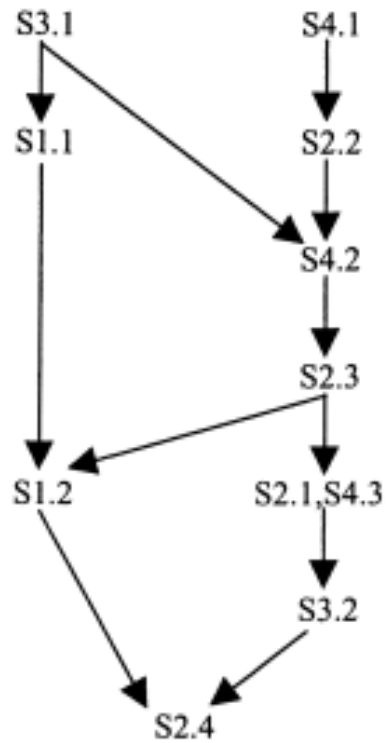
The results of ranking from descending distillation:



The results of ranking from ascending distillation:



The final ranking



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